

Order Flow Imbalance and Price Impact in a Limit Order Book

Multi-Level and Integrated OFI on Real NASDAQ Data,
with Contemporaneous and Out-of-Sample Analysis

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Abstract

Order flow imbalance (OFI) — the net pressure of buy versus sell activity in a limit order book — is among the most robust explanators of short-horizon price movement. Using real level-10 NASDAQ order-book data for AAPL (LOBSTER sample, 21 June 2012; 400,391 events over 6.5 trading hours), I reconstruct OFI from first principles at every book level, and study its relationship to price both contemporaneously and predictively. A single best-level OFI regressor explains 30% of contemporaneous one-second mid-price changes; using all ten levels raises this to 50%, and a single PCA-*integrated* OFI factor recovers nearly all of that gain (48.5%), showing that the deep book carries substantial price information beyond the top of book. The contemporaneous fit strengthens monotonically with the aggregation horizon, reaching $R^2 = 0.74$ (multi-level) at thirty seconds, and the estimated price-impact coefficient is strongly significant under Newey–West inference (HAC $t \approx 11$). In sharp contrast, lagged OFI has almost no out-of-sample predictive power ($R^2 \approx 0.007$ at one second) — the expected signature of an efficient market in which OFI moves price concurrently rather than forecasting it. The study is fully reproducible from the public sample and the code included here.

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Key findings

- Best-level OFI explains 30.1% of contemporaneous 1-second mid-price changes on real AAPL data; all ten levels together explain 50.5%.
- A single PCA-integrated OFI factor captures 48.5% — almost the entire multi-level gain — so the extra information in the deep book is largely one-dimensional.
- Contemporaneous explanatory power rises with horizon: R^2 for multi-level OFI is 0.51 (1s), 0.64 (5s), 0.67 (10s), 0.74 (30s).
- The price-impact coefficient is positive and strongly significant under Newey–West (HAC) standard errors ($t \approx 11$).
- Lagged OFI is *not* predictive out of sample ($R^2 \approx 0.007$, shrinking with horizon), consistent with near-efficiency: OFI accompanies price formation, it does not forecast it.
- The impact is approximately *linear* in OFI (binned $R^2 = 0.94$) and *permanent* (about 131% of the immediate move persists after 20s), and the impact coefficient varies roughly threefold through the trading day.

1 Introduction

When a trade or quote update arrives, it does not move price at random: it moves price in the direction of the net order flow it represents. *Order flow imbalance* (OFI) quantifies that net pressure from changes in the quoted sizes and prices on each side of the book, and a large microstructure literature establishes it as the dominant driver of short-horizon price change [1]. This report studies OFI from first principles on real data, with three questions in view. First, how much of contemporaneous price movement does OFI explain, and does the *deep* book add to what the top of book already tells us? Second, is the relationship stable across time scales and statistically robust once serial correlation is accounted for? Third — the question that separates a describable regularity from a tradeable one — does OFI *predict* future returns out of sample, or only move with price concurrently?

Scope. This is a single-asset, single-day study on the public LOBSTER AAPL sample. That is enough to establish the OFI–price-impact relationship rigorously and to separate contemporaneous explanation from out-of-sample prediction. The multi-asset *cross*-impact extension of [2] — where one asset’s order flow moves another’s price — requires synchronized multi-ticker data and is left as clearly-scoped future work.

2 Data

The LOBSTER sample reconstructs the NASDAQ limit order book for AAPL on 21 June 2012 to ten levels per side, from the official ITCH message stream [3]. Two aligned files are used: a *message* file (one row per event: timestamp, type, order id, size, price, direction) and an *order-book* file (one row per event giving, for each of ten levels, the ask price, ask size, bid price, and bid size). After the opening, the sample contains 400,391 events spanning 34,200–57,600 seconds (09:30–16:00), i.e. 6.5 hours. The mid-price ranges over \$577.48–\$588.18 and the mean quoted spread is \$0.154. Row i of the order-book file is the state of the book immediately after message i , so the two files together give a complete, event-by-event history of the top ten levels.

3 Method

3.1 Order flow imbalance at a single level

Let the best bid be (P_n^b, q_n^b) and best ask (P_n^a, q_n^a) after event n . Following Cont, Kukanov and Stoikov [1], the bid-side order-flow contribution of event n is

$$\Delta W_n^b = \begin{cases} q_n^b, & P_n^b > P_{n-1}^b \quad (\text{bid improves}), \\ q_n^b - q_{n-1}^b, & P_n^b = P_{n-1}^b \quad (\text{same level}), \\ -q_{n-1}^b, & P_n^b < P_{n-1}^b \quad (\text{bid falls}), \end{cases} \quad (1)$$

and symmetrically for the ask side,

$$\Delta W_n^a = \begin{cases} q_n^a, & P_n^a < P_{n-1}^a \quad (\text{ask improves, i.e. falls}), \\ q_n^a - q_{n-1}^a, & P_n^a = P_{n-1}^a, \\ -q_{n-1}^a, & P_n^a > P_{n-1}^a. \end{cases} \quad (2)$$

The per-event order flow imbalance is $e_n = \Delta W_n^b - \Delta W_n^a$, and the OFI over a time bucket $[\tau, \tau + h)$ is the sum of its events,

$$\text{OFI}_\tau = \sum_{n: t_n \in [\tau, \tau+h)} e_n. \quad (3)$$

The sign convention is such that net buying pressure (bid built up, ask consumed) gives $\text{OFI} > 0$, which should push the price up.

3.2 Multi-level and integrated OFI

The same construction applies at each book level $m = 1, \dots, 10$, using that level's price and size, producing $\text{OFI}_\tau^{(m)}$. Deeper levels carry information that the top of book does not — liquidity replenishment and withdrawal below the touch presage where price can go. Rather than treat the ten series as ten free regressors (which risks collinearity), I follow the integrated construction of [2]: standardize each level's OFI, and take the first principal component of their covariance matrix,

$$\text{OFI}_\tau^{\text{int}} = \sum_{m=1}^{10} w_m \widetilde{\text{OFI}}_\tau^{(m)}, \quad w = \text{first eigenvector of } \text{Cov}(\widetilde{\text{OFI}}), \quad (4)$$

where $\widetilde{\cdot}$ denotes standardization. This yields a single, robust depth-weighted factor.

3.3 Price impact and inference

The contemporaneous price-impact model regresses the mid-price change over a bucket (in cents) on the bucket’s OFI,

$$\Delta \text{mid}_\tau = \alpha + \lambda \text{OFI}_\tau + \varepsilon_\tau, \quad (5)$$

with λ the price-impact coefficient. Because bucketed microstructure series are serially correlated and heteroskedastic, ordinary OLS standard errors understate uncertainty; I report Newey–West (HAC) standard errors with Bartlett weights,

$$\widehat{\text{Var}}(\hat{\beta}) = (X^\top X)^{-1} \left(\hat{\Omega}_0 + \sum_{l=1}^L \left(1 - \frac{l}{L+1}\right) (\hat{\Omega}_l + \hat{\Omega}_l^\top) \right) (X^\top X)^{-1}, \quad (6)$$

with $\hat{\Omega}_l$ the lag- l autocovariance of the score $x_i \hat{\varepsilon}_i$ and $L = 10$.

3.4 Out-of-sample predictive test

Contemporaneous explanation does not imply forecasting power. To test prediction honestly I regress the *next* bucket’s return on the current OFI, $\Delta \text{mid}_{\tau+1} = \alpha + \beta \text{OFI}_\tau + \varepsilon$, estimate (α, β) on the first half of the trading day, and evaluate R^2 on the held-out second half. The split is chronological, so no future information enters the fit.

4 Results

4.1 Contemporaneous impact and the value of depth

At one-second buckets (21,029 observations), best-level OFI alone explains $R^2 = 0.301$ of mid-price changes (Figure 1), with a positive, strongly significant impact coefficient ($\lambda = 4.08 \times 10^{-3}$ cents per unit, Newey–West $t = 11.0$). Using all ten levels raises R^2 to 0.505, and the single integrated factor recovers 0.485 — almost the entire gain from one extra degree of freedom (Table 1). Every level carries price information individually (Figure 1; per-level R^2 ranges 0.22–0.30), and the first principal component accounts for 53.7% of the level-OFI covariance, so the deep-book signal is largely one-dimensional — which is exactly why a single integrated factor works so well.

Specification	Contemporaneous R^2 (1s)
Best-level OFI	0.301
Integrated OFI (PCA, 1 factor)	0.485
Multi-level OFI (10 regressors)	0.505

Table 1: Contemporaneous explanatory power at one-second buckets. A single integrated factor captures almost all of the multi-level gain over best-level OFI.

4.2 Robustness across time scales

The relationship strengthens monotonically as the aggregation horizon lengthens (Table 2, Figure 2): multi-level R^2 rises from 0.51 at one second to 0.74 at thirty seconds. This is the expected effect of averaging out event-level microstructure noise, and the ordering best < integrated < multi-level is preserved at every scale.

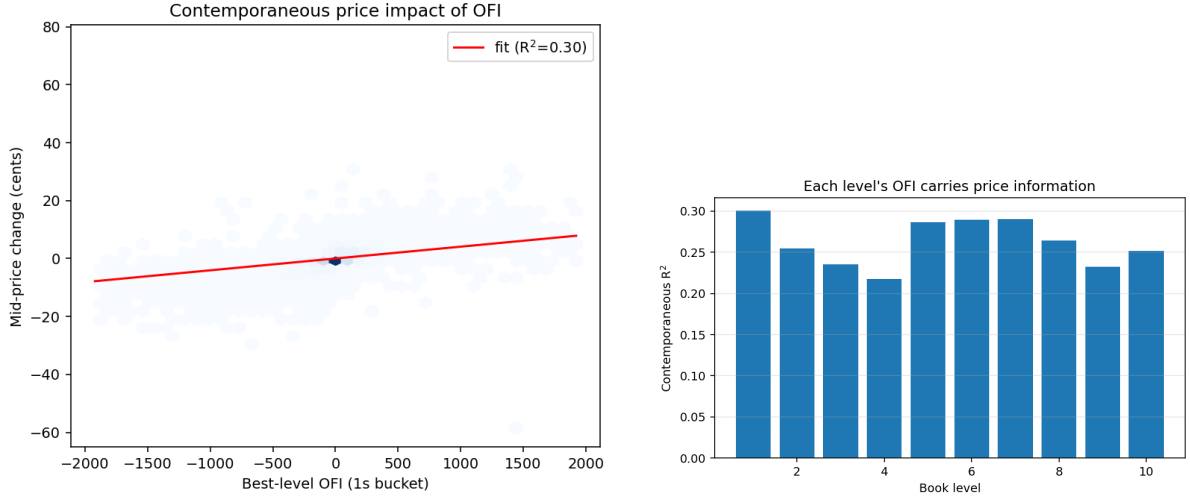


Figure 1: Left: contemporaneous price impact — mid-price change versus best-level OFI (1s buckets), with the fitted line. Right: each book level’s OFI explains price change individually.

Bucket	1s	5s	10s	30s
Best-level	0.301	0.369	0.411	0.480
Integrated	0.485	0.624	0.647	0.707
Multi-level	0.505	0.636	0.668	0.737

Table 2: Contemporaneous R^2 across aggregation horizons.

4.3 Prediction is a different question

When the same OFI is used to forecast the *next* bucket’s return, out-of-sample R^2 collapses to 0.0071 for best-level OFI and 0.0070 for the integrated factor — essentially zero. This is not a failure of the model; it is the correct and expected result. OFI is contemporaneous with price formation: it measures the pressure that is *moving* the price now, not a signal that the market has left un-arbitraged for the next second. The gap between a contemporaneous R^2 of 0.5 and a predictive R^2 of 0.007 is precisely the gap between describing microstructure and forecasting it, and honest evaluation must keep the two apart.

5 Deepening the Analysis

Four further analyses, all on the same real AAPL data, sharpen the picture.

5.1 The impact of OFI is approximately linear

Binning OFI into quantiles and plotting the mean mid-price change per bin (Figure 3) gives a strikingly straight relationship: a linear fit to the binned curve attains $R^2 = 0.94$, higher than a fitted power law ($R^2 = 0.82$; log-log slope 0.35, with mild concavity only at the extremes). This linearity is a known and distinctive feature of *order flow imbalance* — unlike the impact of raw traded volume, which is famously concave (square-root), OFI enters price approximately linearly [1]. It is what makes OFI such a convenient impact variable: a single coefficient λ summarizes the response.

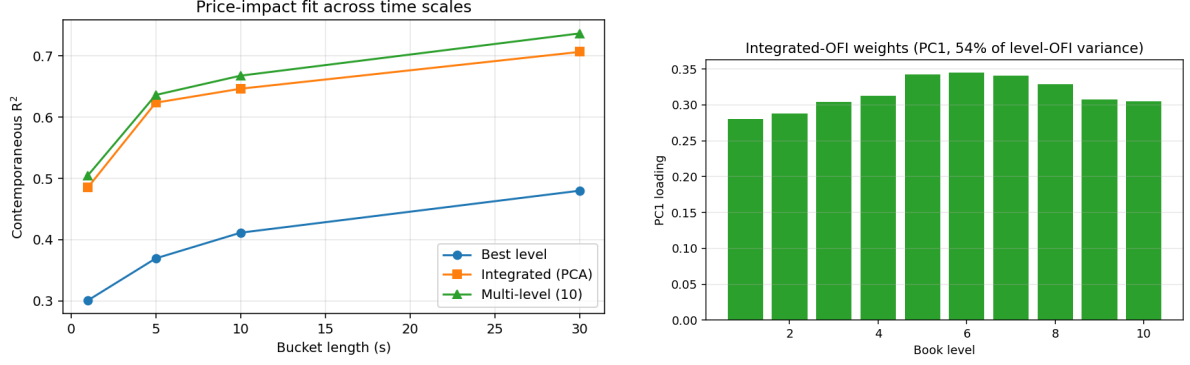


Figure 2: Left: explanatory power versus bucket length. Right: the integrated-OFI weights (first principal component of the ten level OFIs).

5.2 The impact is permanent, not transient

Regressing the cumulative mid-price change from the start of bucket t to the end of bucket $t + \ell$ on OFI_t gives an impact coefficient $\lambda(\ell)$ as a function of horizon (Figure 3). It does not decay: starting at $\lambda(0) = 4.1 \times 10^{-3}$ it is 5.3×10^{-3} after twenty seconds — about 131% of the immediate move. The price does not revert after an OFI shock; if anything it continues to drift in the same direction. This is the signature of *information*, not transient liquidity pressure: order flow imbalance moves the price and the move sticks, consistent with OFI reflecting genuine revisions in valuation rather than temporary inventory effects.

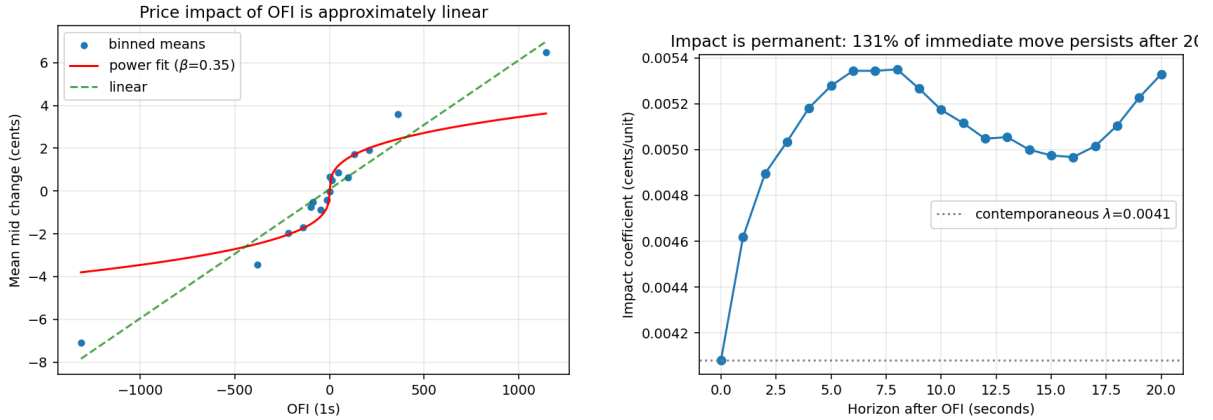


Figure 3: Left: the binned impact curve is close to linear. Right: the impact coefficient does not decay with horizon — the price move is permanent.

5.3 The impact coefficient varies through the day

Estimating λ in thirty-minute bins (Figure 4) shows it is not constant: it ranges from 2.6×10^{-3} to 8.2×10^{-3} cents per unit across the session — a roughly threefold variation, with the largest impact near the open when depth is thin and each unit of imbalance moves price more. A fixed- λ model is therefore a convenience, not a law; impact is state-dependent.

5.4 Predictive power is negligible at every horizon

Extending the out-of-sample predictive test to horizons of one, two, five, and ten seconds (Figure 4) confirms the earlier result at every horizon: the held-out R^2 is 0.006, 0.005, 0.003, and

0.002 respectively — small and shrinking. Whatever information OFI carries is incorporated essentially instantaneously; none of it survives as exploitable predictability a second or more later.

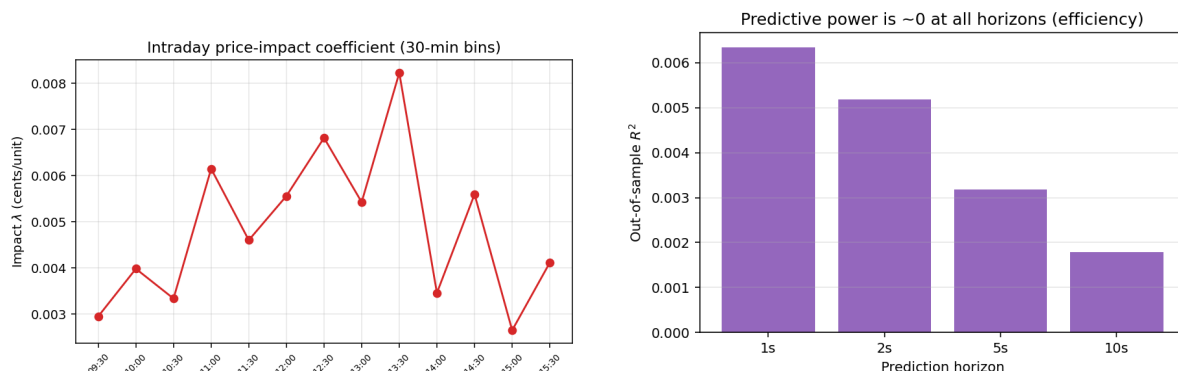


Figure 4: Left: the price-impact coefficient varies about threefold intraday. Right: out-of-sample predictive R^2 is negligible at all horizons.

6 Discussion

Two conclusions stand out. First, the *depth of the book matters*: moving from best-level to all ten levels lifts contemporaneous R^2 by about twenty points, and a single integrated factor captures nearly all of that, so the extra signal is real but low-dimensional. Second, *contemporaneous is not predictive*: the same variable that explains half of concurrent price variation forecasts almost none of it out of sample. For execution and impact modelling — sizing the price move a given order flow will cause — the contemporaneous relationship is exactly what matters; for alpha, OFI at these horizons is close to efficient.

Limitations and extensions. This is one asset on one day, so the estimates are illustrative rather than universal; the clean way to generalize is to repeat across names and dates. The most substantive extension is *cross-impact*: with synchronized multi-asset data, one estimates how asset j 's OFI moves asset i 's price [2], and whether a cross-asset integrated factor improves on per-asset OFI. That requires multi-ticker LOBSTER or vendor data and is the natural next step. Other directions include a non-linear (concave) impact specification and an intraday-varying λ .

7 Reproducibility

All numbers and figures come from a single script (included in Appendix A) that depends only on NumPy, pandas, and Matplotlib. The computation is deterministic. The data is the public LOBSTER AAPL sample; it is not redistributed with the code, which reads it from a local path (see the repository README for the source).

A Source Code: core analysis

```
1 """
2 Order Flow Imbalance and Price Impact in a Limit Order Book
3 =====
4 A from-scratch study of multi-level Order Flow Imbalance (OFI) and its
5 contemporaneous and predictive relationship with price, on real high-frequency
6 NASDAQ limit-order-book data (LOBSTER sample: AAPL, 2012-06-21, level-10 book).
```

```

7
8 Method (after Cont, Kukanov & Stoikov, 2014; multi-level/integrated OFI after
9 Cont, Cucuringu & Zhang, 2023):
10 * Reconstruct best- and deep-level order flow event by event from the book.
11 * Aggregate OFI into fixed time buckets at several time scales.
12 * Contemporaneous price impact: regress mid-price change on OFI; report  $R^2$ 
13 and the impact coefficient with Newey-West (HAC) standard errors.
14 * Compare best-level, multi-level (10), and PCA-integrated OFI.
15 * Predictive test: lagged OFI  $\rightarrow$  next-bucket return, evaluated OUT OF SAMPLE
16 on a chronological train/test split (no look-ahead).
17
18 Data is NOT redistributed here; see README for the public source.
19 Run: python3 ofi_analysis.py data_dir
20 """
21 from __future__ import annotations
22 import sys, os, json
23 import numpy as np
24 import pandas as pd
25 import matplotlib
26 matplotlib.use("Agg")
27 import matplotlib.pyplot as plt
28
29 LEVELS = 10
30 OUT = "outputs"
31
32
33 def load(data_dir):
34     msg = pd.read_csv(f"{data_dir}/msg.csv", header=None,
35                     names=["time", "type", "oid", "size", "price", "dir"])
36     book = pd.read_csv(f"{data_dir}/book.csv", header=None).values.astype(float)
37     return msg["time"].values, book
38
39
40 def level(book, m):
41     o = 4 * (m - 1)
42     return book[:, o], book[:, o + 1], book[:, o + 2], book[:, o + 3] # AskP,AskS,BidP,BidS
43
44
45 def ofi_level(book, m):
46     """Per-event order flow imbalance at level m (Cont-Kukanov-Stoikov)."""
47     ap, asz, bp, bsz = level(book, m)
48     dWb = np.where(bp > np.roll(bp, 1), bsz,
49                 np.where(bp < np.roll(bp, 1), -np.roll(bsz, 1), bsz - np.roll(bsz, 1)))
50     dWa = np.where(ap < np.roll(ap, 1), asz,
51                 np.where(ap > np.roll(ap, 1), -np.roll(asz, 1), asz - np.roll(asz, 1)))
52     e = dWb - dWa
53     e[0] = 0.0
54     return e
55
56
57 def r2(y, yhat):
58     return 1 - ((y - yhat) ** 2).sum() / ((y - y.mean()) ** 2).sum()
59
60
61 def ols(X, y):
62     X1 = np.column_stack([np.ones(len(X)), X])
63     beta, *_ = np.linalg.lstsq(X1, y, rcond=None)
64     return beta, X1, r2(y, X1 @ beta)
65
66
67 def newey_west_se(X1, y, beta, L=10):
68     """HAC (Newey-West) standard errors for OLS coefficients."""
69     n, k = X1.shape
70     resid = y - X1 @ beta
71     XtX_inv = np.linalg.inv(X1.T @ X1)
72     S = (X1 * resid[:, None]).T @ (X1 * resid[:, None])
73     for l in range(1, L + 1):
74         w = 1 - l / (L + 1)
75         Xu = X1 * resid[:, None]
76         G = Xu[l:].T @ Xu[l:-1]
77         S += w * (G + G.T)
78     cov = XtX_inv @ S @ XtX_inv
79     return np.sqrt(np.diag(cov))

```

```

80
81
82 def build_buckets(t, book, E, seconds):
83     bucket = np.floor(t / seconds).astype(int)
84     df = pd.DataFrame(E, columns=[f"ofi{m}" for m in range(1, LEVELS + 1)])
85     df["bucket"] = bucket
86     df["mid"] = (level(book, 1)[0] + level(book, 1)[2]) / 2.0 / 1e4
87     agg = df.groupby("bucket").agg(
88         **{f"ofi{m}": "sum" for m in range(1, LEVELS + 1)}, "mid": ["first", "last"]})
89     agg.columns = [f"ofi{m}" for m in range(1, LEVELS + 1)] + ["mid_first", "mid_last"]
90     agg = agg.reset_index()
91     agg["dmid"] = (agg["mid_last"] - agg["mid_first"]) * 100.0 # cents
92     return agg.iloc[1:-1].reset_index(drop=True)
93
94
95 def integrated(agg):
96     Z = agg[[f"ofi{m}" for m in range(1, LEVELS + 1)]].values
97     Zs = (Z - Z.mean(0)) / Z.std(0)
98     C = np.cov(Zs.T)
99     w, V = np.linalg.eigh(C)
100     pc1 = V[:, -1]
101     if pc1.mean() < 0:
102         pc1 = -pc1
103     return Zs @ pc1, pc1, w[-1] / w.sum()
104
105
106 def main():
107     os.makedirs(OUT, exist_ok=True)
108     data_dir = sys.argv[1] if len(sys.argv) > 1 else "data"
109     t, book = load(data_dir)
110     E = np.column_stack([ofi_level(book, m) for m in range(1, LEVELS + 1)])
111     print(f"events={len(t):,} span={(t.max()-t.min())/3600:.2f}h")
112
113     summary = {"data": "LOBSTER sample AAPL 2012-06-21 (level-10)",
114               "events": int(len(t)), "scales": {}}
115
116     # ---- main analysis at 1s ----
117     agg = build_buckets(t, book, E, 1)
118     y = agg["dmid"].values
119     b1, X1, r2_best = ols(agg[[f"ofi1"]].values, y)
120     se = newey_west_se(X1, y, b1, L=10)
121     _, _, r2_ml = ols(agg[[f"ofi{m}" for m in range(1, 11)]] .values, y)
122     integ, pc1, pc1_var = integrated(agg)
123     _, _, r2_int = ols(integ.reshape(-1, 1), y)
124     per_level = [ols(agg[[f"ofi{m}"]].values, y)[2] for m in range(1, 11)]
125
126     summary["n_buckets_1s"] = int(len(agg))
127     summary["contemporaneous_1s"] = {
128         "best_level_R2": r2_best, "multi_level_R2": r2_ml,
129         "integrated_R2": r2_int,
130         "impact_lambda": b1[1], "lambda_tstat_HAC": b1[1] / se[1],
131         "pc1_var_share": pc1_var, "per_level_R2": per_level}
132     print(f"[1s] best R2={r2_best:.3f} (lambda t_HAC={b1[1]/se[1]:.1f}), "
133         f"multi R2={r2_ml:.3f}, integrated R2={r2_int:.3f}")
134
135     # ---- robustness across time scales ----
136     for s in (1, 5, 10, 30):
137         a = build_buckets(t, book, E, s)
138         yy = a["dmid"].values
139         _, _, rb = ols(a[[f"ofi1"]].values, yy)
140         _, _, rm = ols(a[[f"ofi{m}" for m in range(1, 11)]] .values, yy)
141         ig, _, _ = integrated(a)
142         _, _, ri = ols(ig.reshape(-1, 1), yy)
143         summary["scales"][f"{s}s"] = {"n": int(len(a)), "best": rb,
144                                       "multi": rm, "integrated": ri}
145         print(f"[{s}s] n={len(a)} best={rb:.3f} multi={rm:.3f} integrated={ri:.3f}")
146
147     # ---- predictive, out of sample (chronological split) ----
148     agg["best"] = agg["ofi1"]
149     agg["integ"] = integ
150     agg["y_next"] = agg["dmid"].shift(-1)
151     pred = agg.dropna().reset_index(drop=True)
152     half = len(pred) // 2

```

```

153 def oos(col):
154     tr, te = pred.iloc[:half], pred.iloc[half:]
155     b, _, _ = ols(tr[[col]].values, tr["y_next"].values)
156     Xte = np.column_stack([np.ones(len(te)), te[col].values])
157     return r2(te["y_next"].values, Xte @ b)
158 summary["predictive_oos_1s"] = {"best_level": oos("best"),
159                                "integrated": oos("integ")}
160 print(f"[pred OOS 1s] best={summary['predictive_oos_1s']['best_level']:.4f} "
161       f"integrated={summary['predictive_oos_1s']['integrated']:.4f}")
162
163 # ---- figures ----
164 # 1: OFI vs dmid hexbin (best level)
165 fig, ax = plt.subplots(figsize=(6, 5))
166 m = (np.abs(agg["ofi1"]) < np.percentile(np.abs(agg["ofi1"]), 99))
167 ax.hexbin(agg["ofi1"][m], agg["dmid"][m], gridsize=40, cmap="Blues", mincnt=1)
168 xs = np.linspace(agg["ofi1"][m].min(), agg["ofi1"][m].max(), 100)
169 ax.plot(xs, b1[0] + b1[1] * xs, "r-", lw=1.5, label=f"fit (R2={r2_best:.2f})")
170 ax.set_xlabel("Best-level OFI (1s bucket)"); ax.set_ylabel("Mid-price change (cents)")
171 ax.set_title("Contemporaneous price impact of OFI"); ax.legend()
172 fig.tight_layout(); fig.savefig(f"{OUT}/fig1_impact.png", dpi=140); plt.close(fig)
173
174 # 2: R2 by level
175 fig, ax = plt.subplots(figsize=(6.5, 4))
176 ax.bar(range(1, 11), per_level, color="#1f77b4")
177 ax.set_xlabel("Book level"); ax.set_ylabel("Contemporaneous R2")
178 ax.set_title("Each level's OFI carries price information")
179 ax.grid(axis="y", alpha=0.3); fig.tight_layout()
180 fig.savefig(f"{OUT}/fig2_level_r2.png", dpi=140); plt.close(fig)
181
182 # 3: R2 vs time scale (best/multi/integrated)
183 fig, ax = plt.subplots(figsize=(6.5, 4))
184 scales = [1, 5, 10, 30]
185 ax.plot(scales, [summary["scales"][f"{s}s"]["best"] for s in scales], "o-", label="Best level")
186 ax.plot(scales, [summary["scales"][f"{s}s"]["integrated"] for s in scales], "s-", label="Integrated (PCA)")
187 ax.plot(scales, [summary["scales"][f"{s}s"]["multi"] for s in scales], "^-", label="Multi-level (10)")
188 ax.set_xlabel("Bucket length (s)"); ax.set_ylabel("Contemporaneous R2")
189 ax.set_title("Price-impact fit across time scales"); ax.legend(); ax.grid(alpha=0.3)
190 fig.tight_layout(); fig.savefig(f"{OUT}/fig3_scales.png", dpi=140); plt.close(fig)
191
192 # 4: PC1 loadings
193 fig, ax = plt.subplots(figsize=(6.5, 4))
194 ax.bar(range(1, 11), pc1, color="#2ca02c")
195 ax.set_xlabel("Book level"); ax.set_ylabel("PC1 loading")
196 ax.set_title(f"Integrated-OFI weights (PC1, {pc1_var*100:.0f}% of level-OFI variance)")
197 ax.grid(axis="y", alpha=0.3); fig.tight_layout()
198 fig.savefig(f"{OUT}/fig4_pc1.png", dpi=140); plt.close(fig)
199
200 with open(f"{OUT}/summary.json", "w") as fh:
201     json.dump(summary, fh, indent=2)
202 print(f"\nWrote figures and summary.json to {OUT}")
203
204 if __name__ == "__main__":
205     main()
206

```

B Source Code: extensions

```

1 """Extensions: non-linear impact, impact decay (permanent vs transient),
2 intraday lambda, multi-horizon OOS prediction. Real AAPL LOBSTER data."""
3 import sys, json, numpy as np, pandas as pd
4 import matplotlib; matplotlib.use("Agg"); import matplotlib.pyplot as plt
5 D = sys.argv[1]; OUT = "outputs"; LEVELS=10
6
7 msg = pd.read_csv(f"{D}/msg.csv", header=None, names=["time", "type", "oid", "size", "price", "dir"])
8 book = pd.read_csv(f"{D}/book.csv", header=None).values.astype(float)
9 t = msg["time"].values
10
11 def level(m):

```

```

12     o=4*(m-1); return book[:,o],book[:,o+1],book[:,o+2],book[:,o+3]
13 def ofi_level(m):
14     ap,asz,bp,bsz=level(m)
15     dWb=np.where(bp>np.roll(bp,1),bsz,np.where(bp<np.roll(bp,1),-np.roll(bsz,1),bsz-np.roll(bsz,1)))
16     dWa=np.where(ap<np.roll(ap,1),asz,np.where(ap>np.roll(ap,1),-np.roll(asz,1),asz-np.roll(asz,1)))
17     e=dWb-dWa; e[0]=0.0; return e
18 E=np.column_stack([ofi_level(m) for m in range(1,LEVELS+1)])
19 mid=(level(1)[0]+level(1)[2])/2/1e4
20
21 # 1s buckets
22 bk=np.floor(t).astype(int)
23 df=pd.DataFrame(E,columns=[f"o{m}" for m in range(1,11)]); df["bk"]=bk; df["mid"]=mid
24 agg=df.groupby("bk").agg({**{f"o{m}":"sum" for m in range(1,11)},"mid":["first","last"]})
25 agg.columns=[f"o{m}" for m in range(1,11)]+[f"mf","ml"]; agg=agg.reset_index()
26 agg["dmid"]=(agg["ml"]-agg["mf"])*100
27 # seconds-of-day for the bucket
28 agg["sod"]=agg["bk"]
29 agg=agg.iloc[1:-1].reset_index(drop=True)
30 # integrated OFI
31 Z=agg[[f"o{m}" for m in range(1,11)]].values; Zs=(Z-Z.mean(0))/Z.std(0)
32 w,V=np.linalg.eigh(np.cov(Zs.T)); pc1=V[:, -1]; pc1=pc1 if pc1.mean()>=0 else -pc1
33 agg["oint"]=Zs@pc1
34 best=agg["o1"].values; y=agg["dmid"].values
35
36 res={}
37
38 # ---- 1) Non-linear / concave impact curve (binned) ----
39 q=pd.qcut(agg["o1"],25,duplicates="drop")
40 curve=agg.groupby(q,observed=True).agg(x=("o1","mean"),yv=("dmid","mean")).reset_index(drop=True)
41 # fit power law on positive side: y = a*x^beta
42 pos=curve[(curve.x>0)&(curve.yv>0)]
43 bx=np.log(pos.x.values); by=np.log(pos.yv.values)
44 beta,loga=np.polyfit(bx,by,1)
45 res["impact_exponent_beta"]=float(beta)
46 # linear vs power R^2 on the full binned curve
47 def r2(yt,yh): return 1-((yt-yh)**2).sum()/((yt-yt.mean())**2).sum()
48 lin=np.polyfit(curve.x,curve.yv,1); yhl=np.polyval(lin,curve.x)
49 pw=np.sign(curve.x)*np.exp(loga)*np.abs(curve.x)**beta
50 res["curve_R2_linear"]=float(r2(curve.yv.values,yhl))
51 res["curve_R2_power"]=float(r2(curve.yv.values,pw.values))
52 fig,ax=plt.subplots(figsize=(6,4.5))
53 ax.scatter(curve.x,curve.yv,s=18,color="#1f77b4",label="binned means")
54 xs=np.linspace(curve.x.min(),curve.x.max(),200)
55 ax.plot(xs,np.sign(xs)*np.exp(loga)*np.abs(xs)**beta,"r-",label=f"power fit ($\\beta$={beta:.2f})")
56 ax.plot(xs,np.polyval(lin,xs),"g--",alpha=.7,label="linear")
57 ax.set_xlabel("OFI (1s)");ax.set_ylabel("Mean mid change (cents)")
58 ax.set_title("Price impact of OFI is approximately linear");ax.legend();ax.grid(alpha=.3)
59 fig.tight_layout();fig.savefig(f"{OUT}/fig5_linearity.png",dpi=140);plt.close(fig)
60
61 # ---- 2) Impact response / permanent vs transient ----
62 # lambda(1) = cov(cumulative mid change from start t to end t+1, OFI_t)/var(OFI_t)
63 midend=agg["ml"].values; midstart=agg["mf"].values
64 o=best; vo=o.var()
65 L=20; lam=[]
66 for l in range(0,L+1):
67     # cumulative change from start of bucket t to end of bucket t+1
68     cum=(np.roll(midend,-l)-midstart)*100
69     valid=slice(0,len(o)-l)
70     lam.append(np.cov(cum[valid],o[valid])[0,1]/vo)
71 lam=np.array(lam)
72 res["impact_10"]=float(lam[0]); res["impact_120"]=float(lam[-1])
73 res["permanent_fraction"]=float(lam[-1]/lam[0])
74 fig,ax=plt.subplots(figsize=(6,4.2))
75 ax.plot(range(L+1),lam,"o-",color="#1f77b4")
76 ax.axhline(lam[0],ls=":",color="gray",label=f"contemporaneous $\\lambda$={lam[0]:.4f}")
77 ax.set_xlabel("Horizon after OFI (seconds)");ax.set_ylabel("Impact coefficient (cents/unit)")
78 ax.set_title(f"Impact is permanent: {lam[-1]/lam[0]*100:.0f}% of immediate move persists after {L}s")
79 ax.legend();ax.grid(alpha=.3);fig.tight_layout();fig.savefig(f"{OUT}/fig6_decay.png",dpi=140);plt.close(fig)
80
81 # ---- 3) Intraday variation of lambda (30-min bins) ----
82 agg["halfhour"]=((agg["sod"]-34200)//1800).astype(int)
83 rows=[]

```

```

84 for hh,g in agg.groupby("halfhour"):
85     if len(g)<50: continue
86     b=np.polyfit(g["o1"],g["dmid"],1)
87     yh=np.polyval(b,g["o1"]); rows.append((hh,b[0],r2(g["dmid"].values,yh),len(g)))
88 intr=pd.DataFrame(rows,columns=["hh","lam","r2","n"])
89 res["intraday_lambda_min"]=float(intr.lam.min()); res["intraday_lambda_max"]=float(intr.lam.max())
90 clock=[f"{9+ (34200+hh*1800-34200)//3600 +((34200+hh*1800)%3600)//1800*0:.0f}" for hh in intr.hh]
91 fig,ax=plt.subplots(figsize=(6.5,4))
92 tlabels=[f"{(34200+h*1800)//3600:02d}:{((34200+h*1800)%3600)//60:02d}" for h in intr.hh]
93 ax.plot(range(len(intr)),intr.lam,"o-",color="#d62728")
94 ax.set_xticks(range(len(intr)));ax.set_xticklabels(tlabels,rotation=45,fontsize=7)
95 ax.set_ylabel("Impact  $\lambda$  (cents/unit)");ax.set_title("Intraday price-impact coefficient (30-min bins)")
96 ax.grid(alpha=.3);fig.tight_layout();fig.savefig(f"{OUT}/fig7_intraday.png",dpi=140);plt.close(fig)
97
98 # ---- 4) Multi-horizon OOS prediction ----
99 res["pred_oos"]={}
100 half=len(agg)//2
101 for h in (1,2,5,10):
102     fut=(np.roll(agg["ml"].values,-h)-agg["ml"].values)*100 # cum future return over next h buckets
103     d=pd.DataFrame({"x":agg["oint"].values,"y":fut}).iloc[:len(agg)-h].dropna()
104     tr=d.iloc[:half]; te=d.iloc[half:]
105     b=np.polyfit(tr.x,tr.y,1); yh=np.polyval(b,te.x)
106     res["pred_oos"][f"{h}s"]=float(r2(te.y.values,yh))
107 fig,ax=plt.subplots(figsize=(6,4))
108 hs=list(res["pred_oos"].keys()); vals=[res["pred_oos"][k] for k in hs]
109 ax.bar(hs,vals,color="#9467bd")
110 ax.set_ylabel("Out-of-sample  $R^2$ ");ax.set_xlabel("Prediction horizon")
111 ax.set_title("Predictive power is ~0 at all horizons (efficiency)")
112 ax.grid(axis="y",alpha=.3);fig.tight_layout();fig.savefig(f"{OUT}/fig8_pred_horizon.png",dpi=140);plt.close(fig)
113
114 print(json.dumps(res,indent=2))

```

References

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- [3] Huang, R., & Polak, T. LOBSTER: Limit Order Book System — the efficient reconstructor. Sample data, AAPL 2012-06-21.